
18/20/28-Pin 8-Bit Flash Microcontroller Product Brief

High-Performance RISC CPU:

- Only 49 Instructions
- Operating Speed:
 - DC – 32 MHz clock input
 - DC – 125 ns instruction cycle
- Interrupt Capability with Automatic Context Saving
- 16-Level Deep Hardware Stack with Optional Overflow/Underflow Reset
- Direct, Indirect and Relative Addressing modes:
 - Two full 16-bit File Select Registers (FSRs)
 - FSRs can read program and data memory

Special Microcontroller Features:

- Precision Internal Oscillator:
 - Factory calibrated to $\pm 1\%$, typical
 - Software selectable frequency range from 32 MHz to 31 kHz
- 31 kHz Low-Power Internal Oscillator
- External Oscillator Block with:
 - 4 crystal/resonator modes up to 32 MHz using 4xPLL
 - 3 external clock modes up to 32 MHz
- 4x Phase Locked Loop (PLL)
- Fail-Safe Clock Monitor
- Two-Speed Start-up
- Power-Saving Sleep mode
- Power-on Reset (POR)
- Power-up Timer (PWRT)
- Oscillator Start-up Timer (OST)
- Brown-out Reset (BOR) with Selectable Trip Point
- Extended Watch-Dog Timer (WDT)
- In-Circuit Serial Programming™ (ICSP™)
- In-Circuit Debug (ICD)
- Enhanced Low-Voltage Programming (LVP)
- Operating Voltage Range:
 - 1.8V to 3.6V (PIC16LF1847)
 - 1.8V to 5.5V (PIC16F1847)
- Programmable Code Protection
- Self-Programmable under Software Control

Low-Power Features:

- Standby Current (PIC16LF1847):
 - 30 nA @ 1.8V, typical
- Operating Current (PIC16LF1847):
 - 75 μ A @ 1 MHz, 1.8V, typical
- Low-Power Watchdog Timer Current (PIC16LF1847):
 - 500 nA @ 1.8V, typical

Peripheral Features:

- 15 I/O Pins and 1 Input-only Pin:
 - High current sink/source for LED drivers
 - Individually programmable interrupt-on-change pins
 - Individually programmable weak pull-ups
- Timer0: 8-Bit Timer/Counter with 8-Bit Programmable Prescaler
- Enhanced Timer1:
 - 16-bit timer/counter with prescaler
 - External Gate Input mode
 - Dedicated low-power 32 kHz oscillator driver
- Timer2, 4, 6: 8-Bit Timer/Counter with 8-Bit Period Register, Prescaler and Postscaler
- Two Enhanced Capture, Compare, PWM modules (ECCP):
 - Software selectable time bases
 - Auto-shutdown and auto-restart
 - PWM steering
- Two Capture/Compare/PWM modules (CCP):
 - Software selectable time bases
- Two Master Synchronous Serial Port (MSSP) with SPI and I²C™ with:
 - 7-bit address masking
 - SMBus/PMBus™ compatibility
- Enhanced Universal Synchronous Asynchronous Receiver Transmitter (EUSART):
 - RS-232, RS-485 and LIN compatible
 - Auto-Baud Detect
 - Auto-wake-up on start
- SR Latch (Integrated 555 Timer):
 - Multiple Set/Reset input options
- Analog-to-Digital Converter (ADC):
 - 10-bit resolution
 - 12 channels
- 2 Comparators:
 - Rail-to-rail inputs
 - Power mode control
 - Software controllable hysteresis
- Voltage Reference module:
 - Fixed Voltage Reference (FVR) with 1.024V, 2.048V and 4.096V output levels
 - 5-bit rail-to-rail resistive DAC with positive and negative reference selection
- mTouch™ Sensing Oscillator module
 - 12 channels for button, sensor or slider input
- Data Signal Modulator module
 - Select modulator and carrier sources from various module outputs

PIC16F/LF1847

TABLE 1: PIC16F/LF1847 FAMILY TYPES

Device	Program Memory Flash (words)	Data EEPROM (bytes)	SRAM (bytes)	I/Os	10-bit A/D (ch)	Timers 8/16-bit	EUSART	MSSP	CCP/ ECCP	Cap Touch Channels
PIC16F1847	8192	256	1024	15	12	4/1	Yes	2	2/2	12
PIC16LF1847	8192	256	1024	15	12	4/1	Yes	2	2/2	12

FIGURE 1: 18-PIN PDIP, SOIC PACKAGE DIAGRAM FOR PIC16F/LF1847

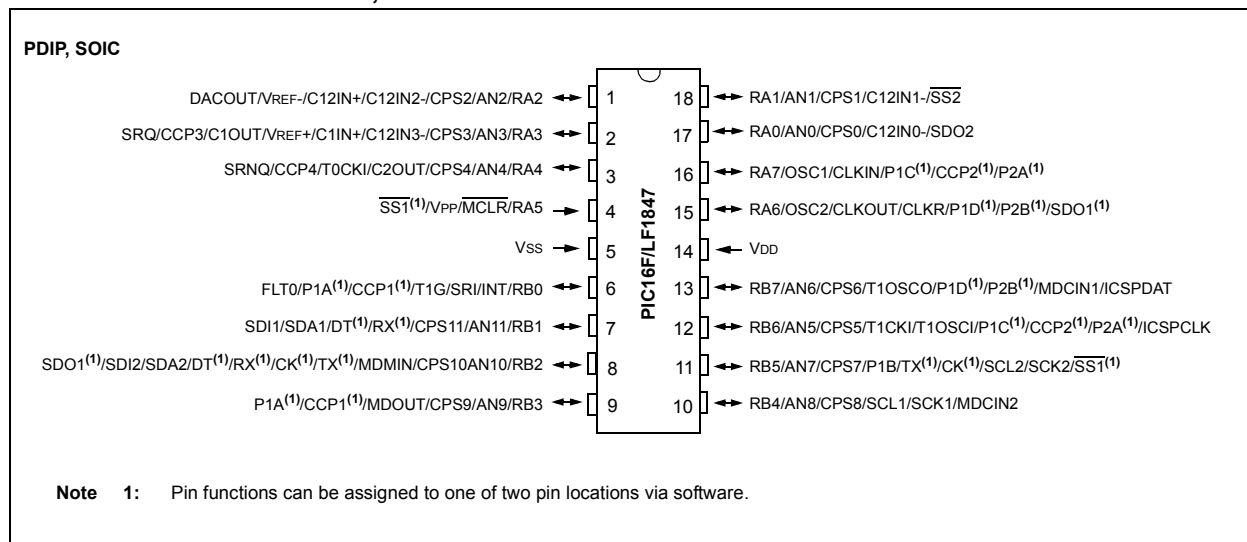


FIGURE 2: 20-PIN SSOP PACKAGE DIAGRAM FOR PIC16F/LF1847

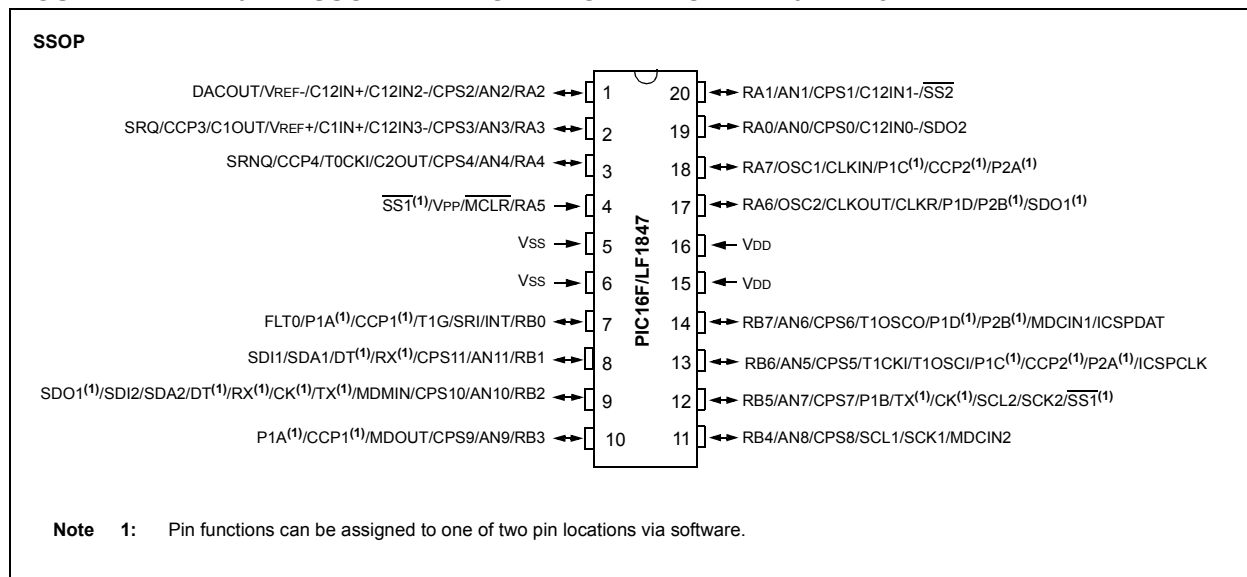
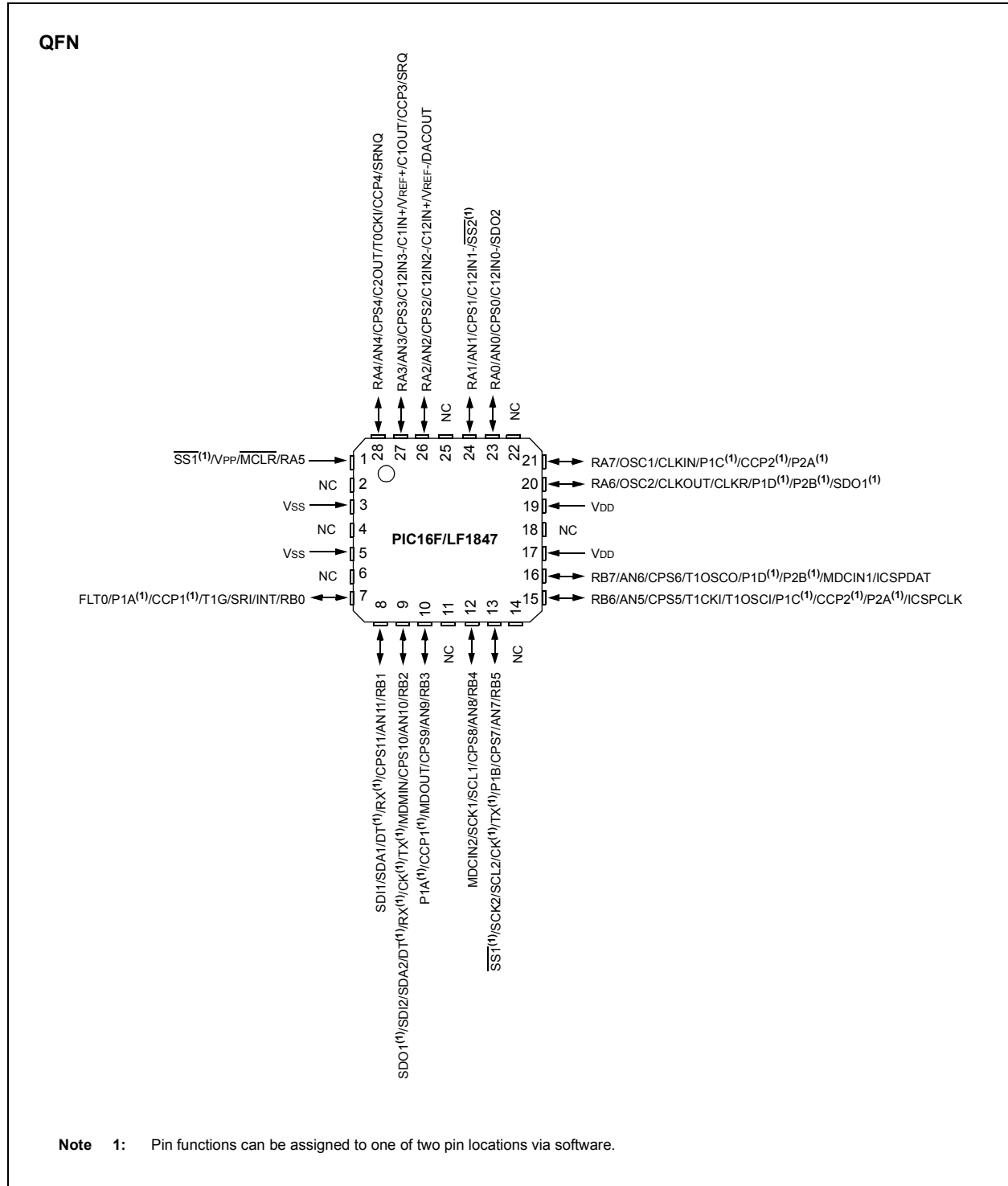


FIGURE 3: 28-PIN QFN PACKAGE DIAGRAM FOR PIC16F/LF1847



PIC16F/LF1847

TABLE 2: PIN ALLOCATION TABLE (PIC16F/LF1847)

I/O	18-Pin PDIP/SOIC	20-Pin SSOP	28-Pin QFN	A/D	Cap Sense	Comparator	SR Latch	Timers	CCP	EUSART	MSSP	Interrupt	Modulator	Pull-up	Basic
RA0	17	19	23	AN0	CPS0	C12IN0-	—	—	—	—	SDO2	—	—	N	—
RA1	18	20	24	AN1	CPS1	C12IN1-	—	—	—	—	SS2	—	—	N	—
RA2	1	1	26	AN2	CPS2	C12IN2- C12IN+	—	—	—	—	—	—	—	N	DACOUT VREF-
RA3	2	2	27	AN3	CPS3	C12IN3- C1IN+ C1OUT	SRQ	—	CCP3	—	—	—	—	N	VREF+
RA4	3	3	28	AN4	CPS4	C2OUT	SRNQ	T0CKI	CCP4	—	—	—	—	N	—
RA5	4	4	1	—	—	—	—	—	—	—	SS1 ⁽¹⁾	—	—	Y ⁽²⁾	MCLR VPP
RA6	15	17	20	—	—	—	—	—	P1D ⁽¹⁾ P2B ⁽¹⁾	—	SDO1 ⁽¹⁾	—	—	N	OSC2 CLKOUT CLKR
RA7	16	18	21	—	—	—	—	—	P1C ⁽¹⁾ CCP2 ⁽¹⁾ P2A ⁽¹⁾	—	—	—	—	N	OSC1 CLKIN
RB0	6	7	7	—	—	—	SRI	T1G	CCP1 ⁽¹⁾ P1A ⁽¹⁾ FLT0	—	—	INT IOC	—	Y	—
RB1	7	8	8	AN11	CPS11	—	—	—	—	RX ⁽¹⁾ DT ⁽¹⁾	SDA1 SDI1	IOC	—	Y	—
RB2	8	9	9	AN10	CPS10	—	—	—	—	RX ⁽¹⁾ DT ⁽¹⁾ TX ⁽¹⁾ CK ⁽¹⁾	SDO1 ⁽¹⁾ SDA2 SDI2	IOC	MDMIN	Y	—
RB3	9	10	10	AN9	CPS9	—	—	—	CCP1 ⁽¹⁾ P1A ⁽¹⁾	—	—	IOC	MDOUT	Y	—
RB4	10	11	12	AN8	CPS8	—	—	—	—	—	SCL1 SCK1	IOC	MDCIN2	Y	—
RB5	11	12	13	AN7	CPS7	—	—	—	P1B	TX ⁽¹⁾ CK ⁽¹⁾	SS1 ⁽¹⁾ SCL2 SCK2	IOC	—	Y	—
RB6	12	13	15	AN5	CPS5	—	—	T1CKI T1OSCI	P1C ⁽¹⁾ CCP2 ⁽¹⁾ P2A ⁽¹⁾	—	—	IOC	—	Y	ICSPCLK ICDCLK
RB7	13	14	16	AN6	CPS6	—	—	T1OSCO	P1D ⁽¹⁾ P2B ⁽¹⁾	—	—	IOC	MDCIN1	Y	ICSPDAT ICDDAT
VDD	14	15, 16	17, 19	—	—	—	—	—	—	—	—	—	—	—	VDD
VSS	5	5, 6	3, 5	—	—	—	—	—	—	—	—	—	—	—	VSS

Note 1: Pin functions can be assigned to one of two pin locations via software.

2: Weak pull-up is always enabled when MCLR is enabled, otherwise the pull-up is under user control.

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